

CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCES

PAPER 02

2 hours 30 minutes

USE ONLY EAST SECONDARY
OFFICE
636-2585
COUNCIL

C.E.S.E.C.

03 MAY 2011 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. Do not open this examination paper until instructed to do so.
2. This paper consists of **SIX** questions, **TWO** from each Module.
3. Candidates **MUST** answer **ALL** questions.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.



MODULE 1

Answer BOTH questions.

1. Figure 1 shows the way in which the number of individuals of two symbiotic species change over time.

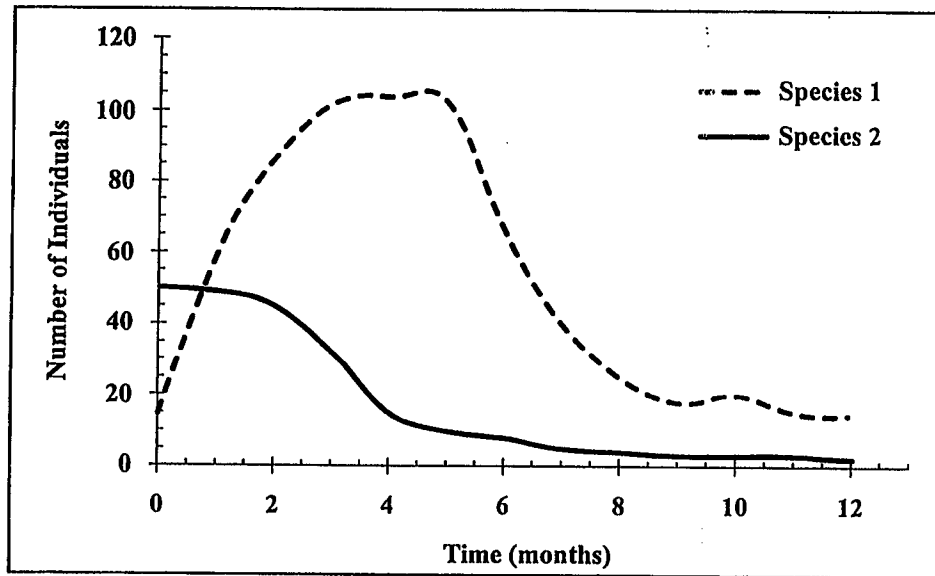


Figure 1. A graph illustrating the way in which the number of individuals of two symbiotic species change over time

- (a) (i) Name the type of symbiotic relationship illustrated in Figure 1. [1 mark]
- (ii) Explain how the symbiotic relationship named in (a) (i) may result in a graph like Figure 1. [4 marks]
- (iii) An analysis of the graph in Figure 1 resulted in the following statement being made.
"The presence of Species 2 is a limiting factor for Species 1".
Is this statement correct? Give ONE reason to justify your answer. [3 marks]
- (iv) After how many months is the difference between the number of individuals in Species 1 and Species 2 at its greatest? Show how you arrived at the answer. [3 marks]
- (v) What is the smallest number of individuals of Species 1? [1 mark]

GO ON TO THE NEXT PAGE

2. (a) You are required to determine the species diversity of some non-mobile (immobile) organisms in a section of a coral reef in the Caribbean. Name and describe the method that you would use to collect the data for calculating the species diversity. [5 marks]
- (b) Figure 3 shows the species data as a stacked bar chart.

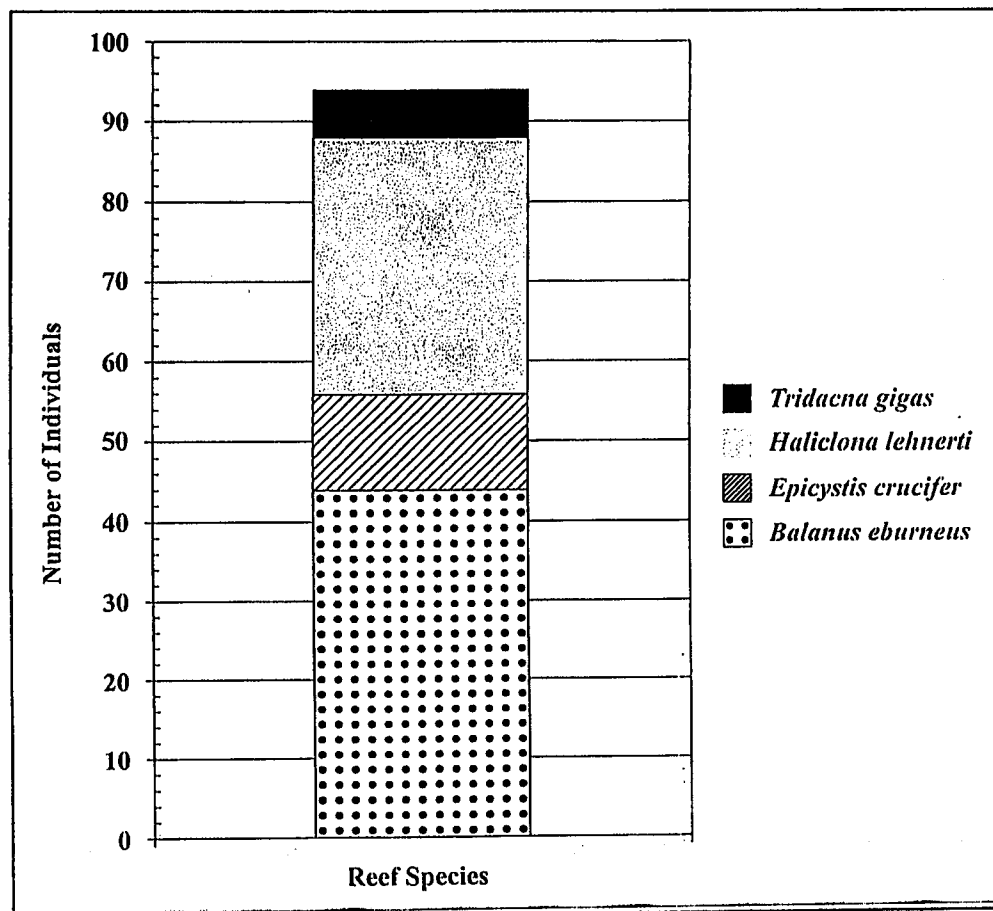


Figure 3. Species diversity data

- (i) Use the data in Figure 3 to determine the number of individuals of EACH species. [4 marks]
- (ii) What percentage of the species is represented by *Haliclona lehnerti*? [1 mark]
- (iii) Calculate the species diversity for the coral reef. [4 marks]
- (c) (i) Define the term 'ecosystem stability'. [2 marks]
- (ii) Explain why a low species diversity may result in poor ecosystem stability. [4 marks]

Total 20 marks

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- (b) Figure 2 shows an outline of an ecological pyramid.

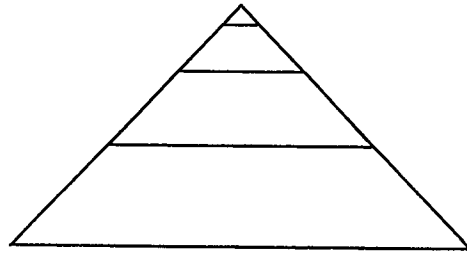


Figure 2. Outline of an ecological pyramid

- (i) Reproduce this outline in your answer booklet and label EACH section with the appropriate organism from the following list:

- Cricket
- Lizard
- Grass
- Mongoose

[4 marks]

- (ii) Using the ecological pyramid in (i) explain the flow of energy in an ecosystem.

[4 marks]

Total 20 marks

MODULE 2

Answer **BOTH** questions.

3. Figure 4 shows age structure diagrams for three countries, Country X, Country Y and Country Z. Study Figure 4 and answer the questions that follow.

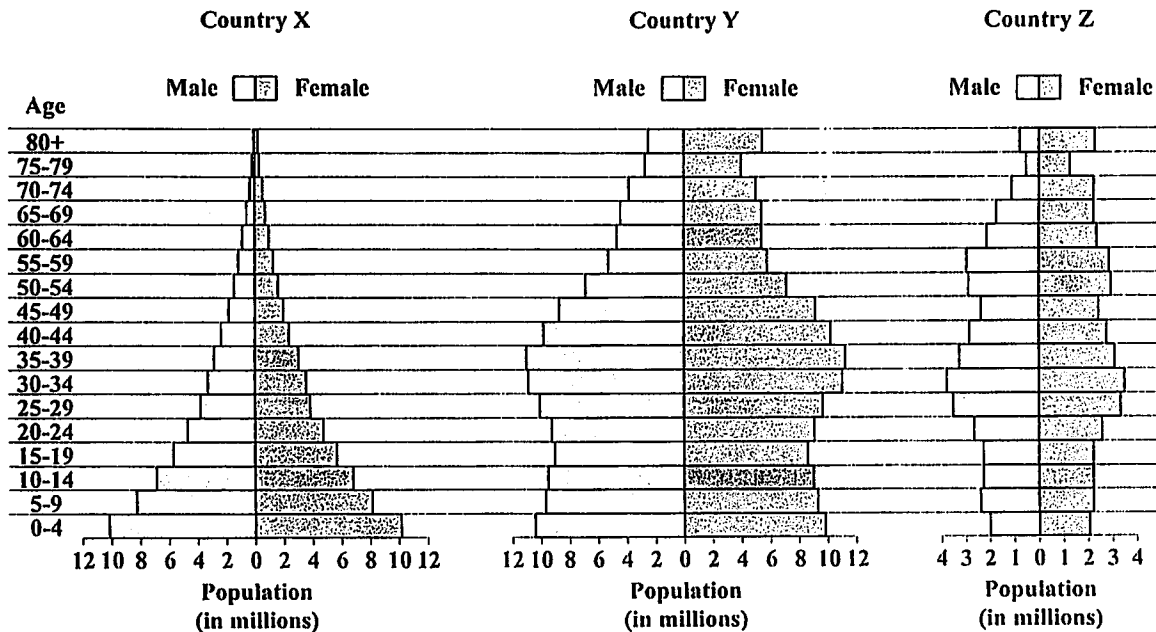


Figure 4. Age structure diagrams for three countries

- (a) Name **THREE** types of population pyramids that are constructed based on mortality and fertility rates. **[3 marks]**
-
- (b) (i) State **ONE** deduction about population growth that can be made for **EACH** of the three countries X, Y and Z. **[6 marks]**
- (ii) Which **ONE** of the three pyramids represents a developing country? Give a reason for your answer. **[2 marks]**
- (c) Discuss how **EACH** of the following factors may affect human population growth rate.
- (i) Age of marriage
 - (ii) Family planning services
 - (iii) Government policy **[9 marks]**

Total 20 marks

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4. (a) Figure 5 shows the percentage of people living in urban and rural areas in three countries, Country A, Country B and Country C. Study the figure and answer the questions that follow.

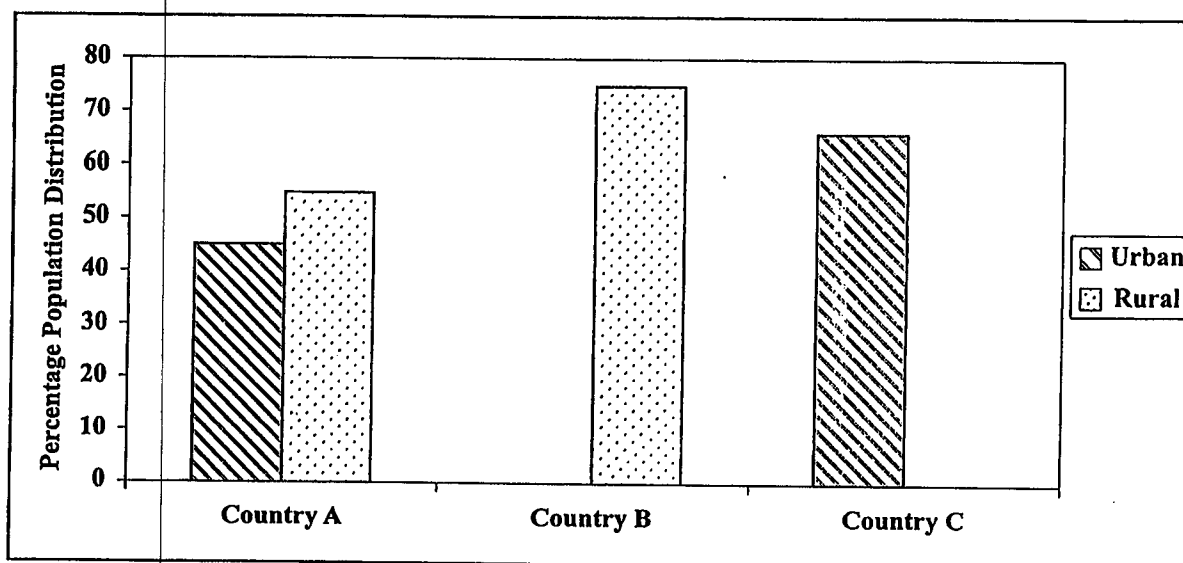


Figure 5. Percentage of people living in urban and rural areas in three countries

- (i) Copy the diagram above in your answer booklet and complete it by inserting the missing bars for Country B and for Country C. Assume that the population for all three countries is comprised of rural and urban only. [2 marks]
- (ii) Compare the rural and urban population distribution in the three countries. [3 marks]
- ~~(iii) Explain why the population growth in Country B may have resulted in the population distribution illustrated in the graph that you have completed. [4 marks]~~
- (b) Outline TWO environmental impacts that are associated with the type of population distribution in Country C. [6 marks]
- (c) (i) State what is meant by the term 'sustainable development'. [2 marks]
- (ii) Explain how an increase in the population may affect a country's ability to achieve sustainable development. [3 marks]

Total 20 marks

MODULE 3

Answer BOTH questions.

5. (a) For any **named** Caribbean country, identify TWO of its natural resources. [3 marks]
- (b) Give FOUR reasons why ONE of the natural resources you identified in (a) above, is important to the named Caribbean country. [8 marks]
- (c) Large areas of Country X are covered in dense forest. In 1975, there was little road access to many of the forested areas, and no governmental regulation of illegal log removal in remote areas of the country. During the period 1975 to 1995, two important events occurred in Country X that affected the illegal tree harvesting trade: a new road was built into the interior of the forested areas by the government and a new environmental policy regulating tree harvesting was put in place. Figure 6 below illustrates the trend in illegal tree harvesting in Country X from 1975 to 1995.

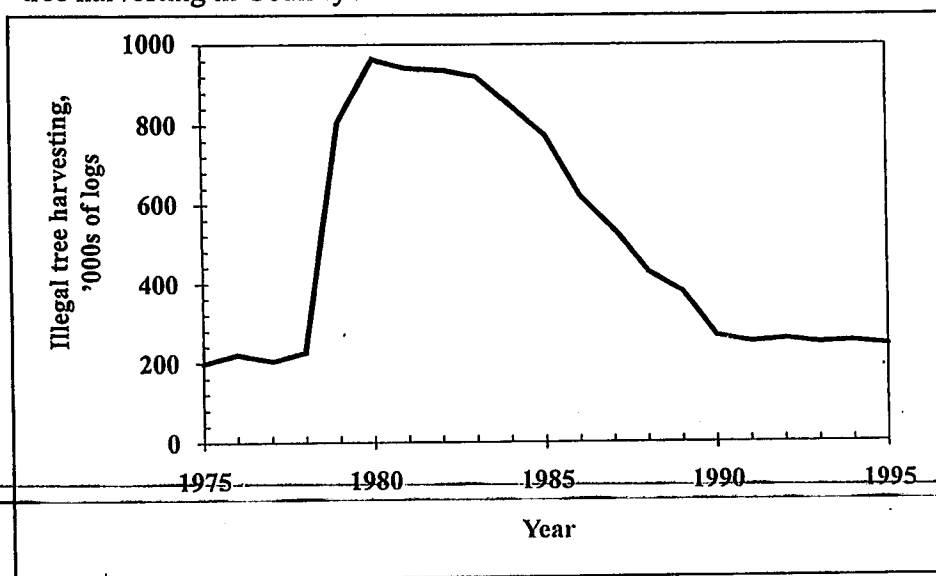


Figure 6. Trend in illegal tree harvesting in Country X from 1975 to 1995

- (i) In what year was the new road built? [1 mark]
- (ii) Explain how the new road might have affected the illegal tree harvesting trade. [3 marks]
- (iii) Explain how the new environment policy would have affected the illegal tree harvesting trade. [3 marks]
- (iv) How long did it take for the illegal tree harvesting trade to return to the level that existed before the new road was built? [2 marks]

Total 20 marks

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6. (a) Give ONE reason why natural resources should be conserved. [1 mark]
- (b) Waste reduction and minimisation has been identified as one method by which natural resources can be conserved. Explain how this practice can result in natural resource conservation. [3 marks]
- (c) Many industries require encouragement to practise waste reduction and minimisation.
- (i) Name TWO measures that can be used to encourage industries to practise waste reduction and minimisation. [2 marks]
 - (ii) Explain how ONE of these measures functions to persuade industries to practise waste reduction and minimisation. [4 marks]
 - (iii) Discuss ONE action that would contribute to the effective use of these measures in the Caribbean. [3 marks]
- (d) The extraction of gravel in quarrying activities can result in significant water pollution due to sedimentation and siltation. To combat this, a series of settling ponds are used to reduce the total suspended solid load leaving the quarry. Figure 7 illustrates the water quality of the quarry effluent before and after each of four settling ponds.

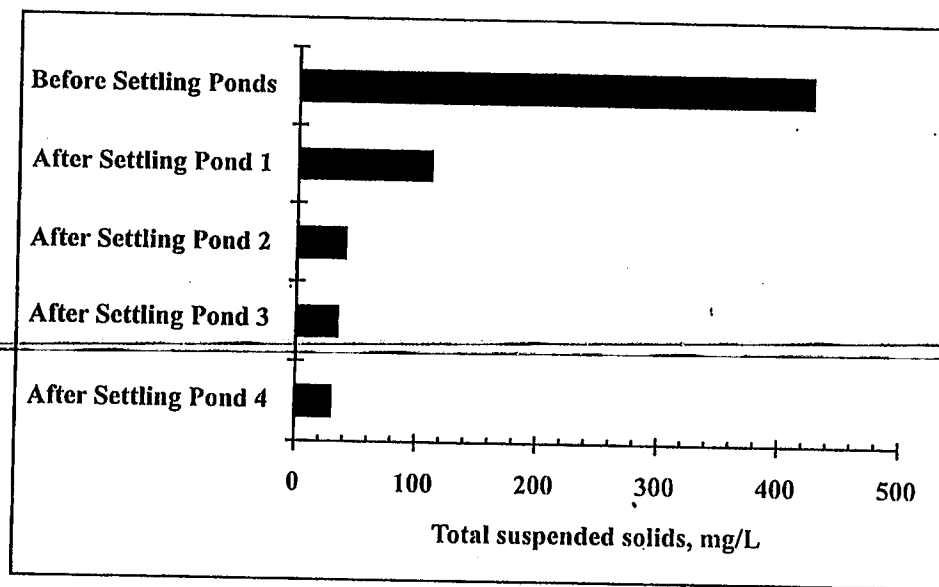


Figure 7. Water quality in quarry effluent before and after settling pond treatment

- (i) What is the percentage reduction of total suspended solids in the quarry effluent after the entire treatment process? [3 marks]
- (ii) What is the percentage reduction of total suspended solids in the quarry effluent after TWO ponds? [2 marks]
- (iii) Are four settling ponds necessary for good pollution control in this quarry? Give ONE reason for your answer. [2 marks]

Total 20 marks

END OF TEST